Refusal as a Safety Mechanism: Consent-Gated Reflective Inference and the Threat Confound

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Submitted to *[AIES / FAccT — select before submission; see §11]*

Abstract

Systems that infer what a person "really" wants from the gap between what they say and what they do face a hazard that is rarely named and, so far as we have been able to establish, has not previously been quantified: the same surface contradiction is produced both by a hidden preference and by a threat response. Telling someone they secretly want something they are afraid of is a specific, foreseeable harm, and no published affect-aware architecture reports its rate.

We present Shadow Mirror, a consent-gated reflective architecture in which refusal is a first-class output, contradiction is held open rather than resolved, and a threat response is represented as a competing hypothesis rather than treated as noise. We evaluate it in simulation against five generative processes with known latent mechanisms, two of which produce identical stated aversion by different routes.

Shadow Mirror misreads a threat response as a hidden desire in **0 of 800 episodes** (point estimate 0.0%; 95% upper bound 0.37% by the rule of three [45]), against **42.9%** [39.5, 46.2] for a stated-versus-behaviour mismatch heuristic and **75.1%** [72.1, 78.1] for a baseline that admits physiological arousal into the action channel. A supervised four-channel classifier, given ground-truth labels our system never sees, matches 0.0% where the confound channels are informative and beats us in the mid-separation regime; we report that comparison in full (§6.5). Guard sensitivity is 100.0% [100.0, 100.0] and specificity 94.4% [93.2, 95.7]. The protection is not free: sensitivity to genuine hidden preference falls from 100.0% to 87.1% [84.9, 89.4], concentrated entirely in high-arousal approach.

An ablation exposes a dependency we report as a design requirement rather than a result. With the avoidance channel unobserved — a common deployment condition — guard sensitivity in a system that imputes the missing channel as zero falls to 0.0% and the misreading rate rises to 31.2% [28.1, 34.4]. We show that treating an absent channel as absent, rather than as evidence of safety, restores sensitivity to 100.0% at the cost of 33 points of specificity, and argue that a reflective system must refuse under partial observation rather than degrade silently.

We close with a tension we cannot resolve. The physiological channel exists only to protect people and demonstrably does so, and it is precisely what brings the system within the EU AI Act's definition of an emotion recognition system (Art. 3(34), 3(39)) and therefore within the Art. 5(1)(f) prohibition on emotion inference in workplaces and education institutions. Removing it removes the legal exposure and reinstates the harm.

This system is a research instrument. It is not a diagnostic device and is not suitable for clinical, educational, employment or other high-stakes deployment.

Keywords: refusal, abstention, consent, affective inference, threat response, harm measurement, EU AI Act, human–AI reflection

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1. Introduction

A large class of deployed systems infers preference from behavioural inconsistency. Recommender systems treat repeated engagement as revealed preference regardless of stated intent. Implicit-measure tools treat reaction-time asymmetries as leanings a person will not admit. Affect-aware interfaces treat physiological arousal as intensity of wanting. In each case the inference has the same structure: *you say one thing, you do another, therefore the second is the truth about you.*

That inference has a failure mode with an established literature in clinical and evolutionary psychology [34–37] and no name at all in machine learning. Approach under stated reluctance is produced by hidden preference. It is also produced by threat: by compliance — by the specific pattern in which a person moves toward something because moving away is unsafe. Arousal is produced by excitement. It is also produced by fear.

The two are not distinguishable from the surface signal. They are distinguishable only with evidence that many deployments do not collect, and the cost of confusing them is asymmetric. Reading a hidden preference as a threat response wastes an opportunity. Reading a threat response as a hidden preference tells a person that their fear was a want — which is, precisely, the thing that a person recovering from coercion has most often already been told.

This paper does three things.

1. **It measures the harm.** We construct generative processes with known latent mechanisms and report the rate at which a system misreads threat as desire — for our architecture and for two baselines that reflect current practice. We are not aware of a prior published rate for this error.
2. **It shows that refusal removes it.** Conditioning the hidden-preference reading on the evidence required to exclude a threat response reduces the misreading rate to zero, and we report the sensitivity and specificity this costs.
3. **It reports a regulatory tension it cannot resolve.** The mechanism that provides the protection is also what places the system inside a prohibition designed to prevent a different harm.

Contribution 1 is the one we would defend most strongly. Design proposals in this space are plentiful; measured harm rates are not.

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2. Related work

2.1 Abstention, selective prediction, and deferral

Declining to answer has a long formal history. Chow [15] gave the optimal error–reject tradeoff for a classifier permitted to abstain; El-Yaniv and Wiener [16] laid the modern foundations of selective classification; Geifman and El-Yaniv extended it to deep networks [17] and then built the reject option into the architecture itself [18]. Confidence-based rejection at the softmax [19] remains the standard baseline, and it is only as good as the confidence, which is why calibration [20, 21] is inseparable from abstention. A parallel literature asks not merely whether to answer but to *whom* to hand the case: learning to defer [22, 23] and the triage view of human effort [24]. Conformal prediction [25, 26, 27] supplies distribution-free coverage guarantees for the set-valued outputs that abstention naturally produces, and Kompa et al. [28] argue the clinical case for communicating that a second opinion is needed.

Our refusal is a different object, and the distinction is the paper's design claim. Every mechanism above abstains because a *confidence* is low — the model is unsure, so it declines, and the threshold is a tunable knob trading coverage against accuracy. Our guard abstains because a *specific alternative explanation cannot be ruled out with the evidence that was collected*. It is not tunable by a confidence parameter, it does not move when the model becomes more certain, and it fires just as readily on a high-confidence reading. A selective

classifier asks "how sure am I?"; the guard asks "which question was never asked?" §6.3 reports what that buys and what it costs — including a specificity cost we do not hide.

2.2 Affect inference and its limits

Affective computing as a field dates to Picard [29]. Its central empirical caution is now well established: Barrett et al. [6] show in an extensive review that emotion cannot be reliably inferred from facial movement, and Gendron et al. [7] that emotion perception is not culturally universal. Calvo and D'Mello [8] survey the methods and their uncertainty.

The ethical critique has kept pace. Stark and Hoey [30] examine the value commitments buried in emotion-AI systems' choice of emotion model; Andalibi and Buss [31] find that people subject to emotion recognition on social media see risks the deploying platforms do not; Roemmich et al. [32] show how emotion-AI hiring services frame organisational problems as technical ones amenable to inference about applicants' inner states.

We treat affective signals as one evidence channel with declared uncertainty and never as a characterisation of an inner state — and §7 argues that even this restrained use falls inside the EU AI Act's definition of an emotion recognition system, which is a problem for us rather than a defence.

2.3 Threat responses, and the vocabulary we borrow

The mechanism this paper is designed to detect — a person complying with something they do not want, under perceived threat — has an established clinical and evolutionary literature. Bracha [34] sets out the acute stress response as a spectrum rather than the textbook fight-or-flight dichotomy, adding freeze and fright. Marx et al. [35] review tonic immobility as an evolved predator defence and its implications for how survivors' non-resistance is misread. Cantor and Price [36] give the most direct antecedent: appeasement as an evolved response to traumatic entrapment, spanning hostage reactions and domestic abuse. Gilbert [37] situates submissive behaviour within social competition and hierarchy.

We are deliberately explicit about one term. The word "fawn", widely used for the appeasement response, does not originate in the peer-reviewed clinical literature: its popular source is Walker [38], a self-published practitioner trade book, and we could locate no refereed source that establishes it as a validated construct. We therefore do not use "fawn" as a clinical category. Where this paper says *threat compliance*, it names a pattern that our generative process encodes and our guard is designed to refuse to overwrite — nothing more. The peer-reviewed constructs above [34–37] describe phenomena in the same region, and we cite them to locate the concern, not to claim our simulation instantiates any of them. §9 restates this as a limitation, because a reader could reasonably conclude we have borrowed more authority than we have earned.

2.4 Trauma-informed computing, and technology-facilitated harm

Chen et al. [33] set out trauma-informed computing as a design orientation, arguing that systems should assume trauma may be present rather than treating it as an edge case. Freed et al. [43] document how intimate partner abusers exploit everyday technology, which is the concrete deployment context in which a system that confidently announces a person's hidden desires becomes an instrument. Our safety argument is downstream of both: the harm we measure is not a wrong prediction in the abstract but a system telling a person that their compliance under pressure was really their own wish.

2.5 Belief revision, human and formal

Bayesian conditioning on conditionally independent observations is commutative [3]; human revision is neither commutative nor minimal, being explanation-guided and order-sensitive [4, 5]. We take the first as a constraint on the arithmetic and the second as a reason to hold contradiction open rather than resolve it.

2.6 Uncertainty, explanation, and reliance

Schäfer et al. [1] show that displaying uncertainty does not by itself produce calibrated reliance; Rosenbacke et al. [2] find in systematic review that explainability can *decrease* clinician trust calibration as easily as increase it. Both inform our position that the honest output of a reflective system is frequently a refusal rather than a better-hedged answer.

2.7 Consent, contextual integrity, and the gap between form and substance

Budin-Ljøsne et al. [12] and Kaye et al. [39] set out dynamic consent: granular, revocable and purpose-locked, designed for exactly the intensive longitudinal collection our system performs. Our ledger (§8) implements that standard and we report where it falls short.

Three critiques bound what any such mechanism can claim. Nissenbaum [40] argues that privacy is governed by norms of information flow appropriate to a context, so that a consent obtained in one context does not license a flow into another. Solove [41] argues that privacy self-management does not scale — that consent transactions are too numerous and too consequentially opaque for individuals to manage meaningfully. Nouwens et al. [42] show empirically that post-GDPR consent interfaces are routinely engineered to produce agreement rather than to elicit it. Wachter and Mittelstadt [44] argue for a right to reasonable inferences, on the ground that data protection law regulates the inputs to inference far more tightly than its outputs — which is precisely the gap a system that infers desire from behaviour occupies.

We take these seriously enough to report an enforcement ratio rather than a consent rate (§8): the question is not how many people agreed but how many refusals the mechanism actually produced.

2.8 Metacognitive architectures

Recent work embeds reflection inside the model: metacognitive reuse [9], meta-memory [10], grounded metacognitive reasoning [11]. Those systems improve what a model knows about its own reasoning. This paper concerns what a system is willing to tell a person about *themselves*, which is a different problem with a different failure mode: being confidently wrong about a model's own chain of thought costs accuracy, while being confidently wrong about a person's motives costs them something else.

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3. The harm, stated precisely

Consider a person who states that they do not want something.

Five mechanisms can produce that statement together with the behaviour that follows. Three of them are genuine preferences. Two are not:

Mechanism	Stated	Behaviour	Arousal	Avoidance	Is it a preference?
Explicit aversion	low	withdraws	low	low	yes — they do not want it
Latent approach	low	approaches	low	low	yes — they want it and have not said so
Latent approach, aroused	low	approaches	**high**	low	yes — excitement
Threat response	low	withdraws	**high**	**high**	**no** — fear
Threat compliance	low	**approaches**	**high**	moderate	**no** — complies under threat

The fifth row is the dangerous one. **Threat compliance and latent approach are indistinguishable on the stated and behavioural channels alone.** Both show stated aversion and approach behaviour. A system that reads only those two channels — which describes most deployed preference inference — cannot

tell them apart, and will call the compliance case a hidden want.

The third and fourth rows are the second trap. Excitement and fear both raise arousal. A system that treats arousal as intensity-of-wanting will call the threat case a strong want.

We take as our target metric the rate at which a system outputs `latent_approach` — "you want this even though you say you don't" — when the true generating mechanism was a threat process. We call it the **false hidden-desire rate**.

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4. Design

Four commitments, each with a code-level enforcement point.

C1. Refusal is a first-class output. The system emits `unresolved_approach` — a named state, not a low-confidence label — when behaviour approaches under stated reluctance and the evidence needed to exclude a threat response was not collected. It also refuses entirely below a minimum of paired evidence.

C2. Absent is not zero. An unobserved channel is recorded as unobserved. It is never imputed to a value, because imputing arousal or avoidance to zero is imputing *evidence of safety*. §6.2 measures what happens when this commitment is dropped.

C3. Threat is a competing hypothesis, not a filter. `threat_response` occupies the same hypothesis space as the preference readings and is scored against them, rather than being applied as a post-hoc suppression rule.

C4. Contradiction is held, not resolved. Supporting and contradicting evidence are both retained as typed edges; a hypothesis carrying both is reported as contested rather than collapsed to the later write. Belief accumulates in log-odds so that revision is order-invariant and exactly retractable — clamping the state destroys retractability, and a system that cannot undo an inference cannot honour a withdrawal of consent.

The output is a distribution over named states with an explicit uncertainty, or a refusal. There is no scalar score anywhere in the API.

4.4 A request the architecture declines

Everything reported so far concerns a threat confound we generated. This section reports a refusal that was not generated: a real request, from the person who commissioned this work, declined for stated reasons that a reader can check by running the code.

The request. To add a feature that would "determine and understand who people probably are according to their will, desire and need — where they come from, how mature they are, what they still need to be mature enough, what their eventual purpose probably is". It arrived with a qualification: *"I would not say judge people."*

That qualification did most of the work. It signals an intuition that something in the request was wrong without identifying which part, and the architecture's job here was to make the split legible rather than to accept or refuse wholesale.

The split. Five asks, separating on whether a validated construct exists and who supplies it.

Ask	Disposition	Ground
will, desire, need	**built**	validated self-report exists; the person supplies it
where they come from	**declared only**	inference of special-category data (GDPR Art. 9; PDPL Art. 1); on biometric input, prohibited by AI Act Art. 5(1)(g)

Ask	Disposition	Ground
how mature they are	**refused as a rank**	the rank is the construct, not a summary of it
what they still need	**refused**	a directive about identity
eventual purpose	**refused**	no validated construct exists; any output would be invented

The maturity row is the one that took work. We looked for an honest way to report a developmental construct without collapsing a person into a single number, and found that the instruments which measure it will not permit that. Loevinger's sentence completion test [51] yields a Total Protocol Rating; Kegan's subject-object interview [52] yields one position on one ordered scale, defended in writing; Cook-Greuter's MAP reports a "Center of Gravity" (we located no independent peer-reviewed validation of the current instrument). In each the ordered position *is* the theory's claim, so a profile drawn from one would misrepresent it. We therefore dropped the construct rather than softening it, and substituted instruments that never claimed a stage — Ardelt's three-dimensional wisdom scale [53], whose own published criterion is conjunctive across dimensions rather than aggregate, and the Modified Erikson Psychosocial Stage Inventory [54], which measures attributes *associated with* stages rather than attainment of them.

What replaces the refused asks. Not silence. "What do they still need" becomes `open_questions()`, which returns what the model does not know, phrased as invitations — and is guarded by a test that fails if the phrasing drifts into *lack*, *deficit*, *should* or *must*. "Their eventual purpose" becomes the person's own answers to the Meaning in Life Questionnaire [55], which measures presence of and search for meaning; needs are measured by the Basic Psychological Need Satisfaction and Frustration Scale [56]. The distinction is not cosmetic: one reports what the person said, the other tells them what they are for.

Two commitments that cost functionality. No item text ships with the software; the registry carries scoring rules, citations, licence status and caveats, and the researcher supplies the licensed instrument. And an unconfirmed licence is a refusal rather than a warning — of six otherwise usable instruments, only two carry quoted published permission statements, so the other four cannot be administered until written permission is on file. Confirming permission requires written evidence of who granted it and when, not a boolean, for the same reason the safety exception in §4 does.

Why this belongs in a paper about refusal. The safety result of §6.1 measures a refusal we designed against a confound we simulated. This one was demanded by a real person, and the architecture's answer was to decline four fifths of it and say why in terms the person can read. A reviewer sceptical that our refusals are load-bearing rather than decorative can run `check_xi_declared_only` against a model in which the refusal has been removed, and watch it fail.

5. Method

Generative processes. Five processes, defined by target values on four channels (stated, action, arousal, avoidance), with observations drawn from Beta distributions around those targets. Ground truth is the generating process, which exists independently of the inference model.

Misspecification control. The inference model assumes each latent state sits at a fixed corner of a signal space. Generating from exactly those corners would test self-consistency rather than validity, so every generating centre is displaced by δ in a random direction. We report $\delta = 0.00$ and $\delta = 0.10$.

Baselines, both implemented and executed at benchmark time:

- **Mismatch-only** — stated versus behaviour, no confound channels. Represents systems that infer preference from behavioural inconsistency alone.

- **Arousal-as-desire** — folds physiological arousal into the action channel. Not a straw man: treating autonomic arousal as an approach signal is documented practice in affect-aware systems.

Statistics. n = 400 episodes per process per condition, 6 observation rounds each. All rates carry 2,000-resample percentile bootstrap intervals [46]. The suite is deterministic: two full runs produce byte-identical output.

What is not measured. No human participants were involved at any stage. Every result below is a statement about simulated signal streams with known generating mechanisms, and none is a statement about people. [NO DATA YET — human evaluation]

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6. Results

6.1 The safety result

δ = 0.10, both confound channels observed, n = 400 per process:

Method	False hidden-desire rate	True hidden preference detected
Shadow Mirror	**0.0% [0.0, 0.0]**	87.1% [84.9, 89.4]
Mismatch-only baseline	42.9% [39.5, 46.2]	100.0% [100.0, 100.0]
Arousal-as-desire baseline	75.1% [72.1, 78.1]	89.5% [87.4, 91.8]

Guard sensitivity 100.0% [100.0, 100.0]; specificity 94.4% [93.2, 95.7]; AUC of the threat score across all five processes 1.000. At δ = 0.00 the figures are 0.0% [0.0, 0.0], 47.4% [44.0, 50.9] and 75.8% [72.8, 78.6], with specificity 97.4% [96.5, 98.3].

Per process, δ = 0.10, the false hidden-desire rate:

True process	Shadow Mirror	Mismatch-only	Arousal-as-desire
`threat_response`	**0.0%**	0.0%	50.5%
`threat_compliance`	**0.0%**	**85.8%**	99.8%

The compliance row is the finding. A system reading stated-versus-behaviour alone gets the withdrawal-type threat response right — it looks like aversion — and gets the *compliance* case wrong **85.8%** of the time. That is the case where a person approaches because approaching is safer, and it is the case where being told "you actually want this" does the most damage.

The cost, stated plainly. Sensitivity to genuine hidden preference is 87.1% against the baseline's 100.0%. The entire loss is in the high-arousal approach case, detected 74.2% [70.0, 78.5] of the time, where excitement is mistaken for fear. Approximately 13 points of sensitivity buys the elimination of the misreading. We regard that as the correct side of the trade for an instrument that hands its output to a person about themselves, but it is a trade and not a free lunch.

6.2 The ablation: what "absent is not zero" is worth

Deployments frequently lack an avoidance measure. We ablate that channel and compare two treatments of the missing data.

	Imputes missing channel as zero	Records it as absent (ours)
Guard sensitivity	**0.0% [0.0, 0.0]**	**100.0% [100.0, 100.0]**
False hidden-desire rate	**31.2% [28.1, 34.4]**	**0.0% [0.0, 0.0]**

	Imputes missing channel as zero	Records it as absent (ours)
…on `threat_compliance`	62.5% [57.8, 67.2]	0.0% [0.0, 0.0]
Specificity	100.0%	67.0% [64.4, 69.8]
True hidden preference detected	99.2%	0.0% — the system refuses
AUC, fear vs excitement	—	0.835

Imputing a missing channel to zero is imputing evidence of *safety*, and it silently disables the guard: sensitivity falls to zero and the compliance case is misread 62.5% of the time, with nothing in the output indicating that the protection is inactive.

Recording it as absent produces the opposite failure and the better one. The system stops asserting hidden preference entirely — sensitivity to genuine preference goes to zero because it declines to make the call — and specificity falls to 67.0% because arousal alone cannot separate fear from excitement (AUC 0.835), so the caution fires on excitement too.

This is the paper's central design claim. Under partial observation a reflective system should refuse, not degrade; and the refusal must be visible in the output, not implicit in a number that happens to be low.

6.3 Refusal calibration

Refusal is only defensible if it is calibrated. n = 300 per condition:

Condition	Behaviour
Evidence genuinely insufficient	refuses **100.0% [100.0, 100.0]**
Evidence sufficient and clear	over-refuses **0.0% [0.0, 0.0]**
Evidence ample but ambiguous	does not refuse; reports uncertainty **0.405 [0.391, 0.419]** against **0.021 [0.020, 0.021]** on clear evidence
Evidence present but expired (decade-old, 30-day half-life)	infers **0.0% [0.0, 0.0]** of the time

The system refuses when it should, does not refuse when it should not, and separates ambiguous from clear evidence by a factor of nineteen in reported uncertainty.

6.4 Two supporting results

Ambivalence is a state, not a residual. "You want and do not want this at once" is a claim about a person. Computed from dispersion alone it fires on anything spread out: 95.5% of pure measurement noise and 98.8% of monotonic drift — that is, of someone who has *made up their mind* — read as ambivalence, for a precision of 34.0%. Gating on lag-1 autocorrelation, which distinguishes alternation (−0.844) from drift (+0.517) and noise (−0.120) where dispersion cannot (0.357 / 0.245 / 0.257), raises precision to **97.6%** with recall unchanged at 100.0%.

Inquiry is information-driven. When the system asks for more evidence, it selects the observation that would most reduce uncertainty. Normalised regret against an information-theoretic oracle is **0.161 [0.138, 0.187]** with a belief-gap heuristic at 0.579 [0.547, 0.612] and random selection at 0.550 [0.515, 0.584] — the heuristic being indistinguishable from random.

6.5 Adversarial robustness: two checks a sceptical reader should demand

Two objections are strong enough that answering them in prose would be evasive.

"Your simulator is your theory." The generative processes were defined so that threat and approach differ on arousal and avoidance, and the guard keys on arousal and avoidance. A 0.0% rate under those conditions risks being a tautology, and the AUC of 1.000 is the tell. So we swept the *separation* between the two processes on the confound channels, from full (as in §6.1) to none, at $n = 300$ per process per level.

Separation	AUC	Shadow Mirror	Four-channel classifier	Mismatch-only
1.00	1.000	**0.0%** [0.0, 0.0]	**0.0%** [0.0, 0.0]	94.3% [91.7, 96.7]
0.75	1.000	**0.0%** [0.0, 0.0]	**0.0%** [0.0, 0.0]	95.3% [92.7, 97.7]
0.50	1.000	2.3% [0.7, 4.0]	**0.0%** [0.0, 0.0]	96.7% [94.7, 98.7]
0.25	0.988	29.7% [24.7, 35.0]	**5.0%** [2.7, 7.7]	97.3% [95.3, 99.0]
0.00	0.491	76.7% [71.7, 81.3]	82.0% [77.7, 86.3]	96.3% [94.0, 98.3]

The headline is not a knife-edge: it survives to roughly half the original channel separation. But it is **entirely contingent on the confound channels carrying signal**, and at zero separation every method fails, as it must — the classes are then identical on every observable, and any method reporting 0.0% would be reporting a bug.

"Your baselines are straw men." They partly were. Mismatch-only ignores channels that are available; arousal-as-desire is a mistake rather than a method. The table above therefore adds a baseline a competent engineer would build: a threshold classifier over the same four channels, with its threshold **fitted on a held-out labelled split**.

It matches us at high separation and **beats us at separation 0.25 — 5.0% against our 29.7%.** We report this because it is true and because the reason matters: our guard uses a fixed, pre-specified threshold, while the classifier was handed ground-truth labels for "was this a threat response".

Those labels do not exist in deployment. Nobody can tell a running system, for a given person and episode, which mechanism produced their behaviour; that is the entire difficulty. The supervised baseline is therefore an *upper bound on what any labelled method could achieve*, not an available alternative — and the gap between it and our unsupervised guard at separation 0.25 is a real, quantified cost of not having supervision, which we would rather state than omit.

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7. The regulatory tension

We report this because it is unresolved, and because the shape of it is more general than this system.

The EU AI Act defines biometric data as *"personal data resulting from specific technical processing relating to the physical, physiological or behavioural characteristics of a natural person"* (Art. 3(34)) [13]. Unlike GDPR Art. 4(14) [14], **it does not require unique identification**. A heart-rate-like arousal proxy is therefore biometric data under the Act while identifying nobody.

An *emotion recognition system* is one *"for the purpose of identifying or inferring emotions or intentions of natural persons on the basis of their biometric data"* (Art. 3(39)). Our guard infers a threat response — an affective and intentional state — partly on the basis of that arousal channel.

Art. 5(1)(f) prohibits *"the use of AI systems to infer emotions of a natural person in the areas of workplace and education institutions, except where the use of the AI system is intended to be put in place or into the market for medical or safety reasons."* Applicable since 2 February 2025.

So: **the channel that exists only to protect people, and that §6.2 shows does protect them, is what brings the system within a prohibition written to prevent a different harm.** Remove it and the exposure largely disappears; remove it and the false hidden-desire rate returns to 31.2%.

The safety exception is available and, in our reading, applicable: the channel's sole documented purpose in code, tests and specification is to prevent a specific measured harm, and it is used for nothing else. We state that as an argument rather than a conclusion. We are aware of no case law interpreting the exception for a guard of this kind, and a system whose safety mechanism is itself an emotion-inference system is a case the Act does not obviously anticipate.

We think this generalises. Any protective mechanism that works by observing more about a person will sit in tension with rules that limit observation. Naming that tension seems more useful than choosing a side of it in a paper.

7.1 A second regime, with no research exemption

The EU analysis above is the one the literature rehearses. It is not the only regime a system like this must survive, and the contrast is instructive enough to report.

This work is conducted under Egyptian jurisdiction, where two instruments matter. Law No. 214 of 2020 on Clinical Medical Research [47] establishes a national ethics architecture — a Supreme Council, registered institutional committees, criminal penalties reaching ten years where a participant dies. It does not reach us. Article 1 defines its subject as studies assessing "therapeutic, pharmaceutical, surgical, nutritional, preventive or diagnostic interventions", and Article 24 confines lawful research to entities registered with the Ministry of Health. Behavioural, psychological and computing research falls outside its scope, a gap noted in the empirical research-ethics literature [49]. Egypt's Supreme Council of Universities issued a national academic-integrity and research-ethics guide in January 2026 whose third book addresses behavioural and social research committees [50], but it carries no decree number and no binding-force clause, and whether it reaches private universities is unspecified. We classify it as guidance rather than law.

Personal Data Protection Law No. 151 of 2020 [48] does reach us, and reaches further than the GDPR in three respects that matter here. Its definition of personal data expressly includes data determining "psychological … identity". Its sensitive-data category (Art. 1) covers "psychological, mental, or physical health, or genetic, biometric or financial data", and Article 12 prohibits processing that category without a licence from the Personal Data Protection Centre *in addition to* written explicit consent, on penalty of imprisonment and a fine up to EGP 5,000,000. And — the point we want to put in front of this venue — **it contains no scientific-research exemption whatsoever**. The GDPR's research accommodations at Arts. 5(1)(b), 9(2)(j) and 89 have no Egyptian counterpart. We searched the Arabic text of both the law and its executive regulations (Ministerial Decree No. 816 of 2025, in force 2 November 2025, with a compliance deadline of 31 October 2026): the words for "research", "studies" and "university" do not appear. Article 5(6) relaxes purpose limitation for not-for-profit statistical or educational processing, but it does not disapply consent or the Article 12 licence.

The consequence is worth stating plainly, because it inverts a common assumption. A researcher in a jurisdiction with a *weaker* national human-subjects law can face a *stricter* data-protection constraint on exactly the physiological and psychological channels this architecture consumes, with no research carve-out to fall back on. Our design response is the one already described: refuse the biometric stream rather than seek a basis for it (§4), and treat anonymisation rather than pseudonymisation as the only route out of scope, since a linkable participant identifier keeps the data personal.

This is a reading of statutory text, not legal advice, and it has not been reviewed by counsel (§9). We report it because the fairness literature's regulatory analysis is overwhelmingly EU- and US-centred, and a system designed only against the regimes that grant research exemptions will be designed wrong for the places that do not.

8. Consent, and the honest enforcement ratio

The system's other-person mode requires granular, per-stream, purpose-locked, revocable consent before any datum is recorded. Streams a subject cannot notice about themselves — implicit measures, physiological arousal, micro-avoidance — cannot be granted under a blanket authorisation; each requires its own acknowledgement. Every stream declares in plain language both what it records and what it *may reveal*. No grant, no observation: the engine raises rather than records.

Assessed against GDPR Art. 4(11) and Art. 7, the implementation meets five criteria and exceeds two (unbundleable non-obvious streams, and may-reveal disclosure). It falls short on two: there is no append-only consent record, so a controller cannot demonstrate what consent was true at a past date (Art. 7(1)); and withdrawal, while a single command, produces no receipt to the subject (Art. 7(3) parity). Both are reported here rather than fixed before publication, because the enforcement ratio is the point.

That ratio: of twelve documented prohibitions, five are enforced by a mechanism and covered by a test; seven are policy only. Among the seven is the one prohibition that is currently unlawful to breach in the EU — emotion inference in a workplace or education institution — for which no code mechanism exists. We consider a deployment-context declaration that refuses to enable the physiological stream in institutional contexts, and we have not implemented it.

We suggest that papers proposing ethical architectures should print this ratio. A list of prohibitions communicates intent; the fraction of them a machine will actually refuse communicates the safeguard.

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9. Limitations

No human subjects. Nothing here establishes any effect on human reasoning, trust, or wellbeing, and no such claim is made. [NO DATA YET — human evaluation]

Simulation is not the world. Our generative processes encode a theory of how threat and preference differ in observable signal. If that theory is wrong, the measured rates are wrong in the same direction for every method compared, including the baselines. We regard the *ordering* of the methods as more robust than the absolute rates, and §6.5 measures how far that ordering survives as the theory's key assumption is weakened.

A point estimate of zero is not a guarantee of zero. Across 800 threat episodes we observed no misreadings. The 95% upper bound implied by the rule of three [45] is 0.37%, and that — not 0.0% — is the number a deployment decision should use.

The clinical vocabulary is borrowed, and one term is not clinical at all. We use "compliance" and "threat response" to name a pattern of approach behaviour under low stated desire and high arousal. There is a genuine peer-reviewed literature in this region — appeasement under traumatic entrapment [36], tonic immobility as an evolved defence [35], the acute stress response as a spectrum rather than a dichotomy [34], submissive behaviour in social hierarchies [37] — and we cite it in §2.3 to locate the concern. We do not claim our generative process instantiates any of those constructs, and no clinical validation has been performed.

One term deserves separate treatment. The word "fawn" is in wide circulation for the appeasement response, and we used it in earlier drafts as though it were a clinical category. It is not. We traced it to Walker [38], a self-published practitioner trade book, and could locate no refereed source establishing it as a validated construct. We have therefore removed it from every load-bearing claim in this paper and retain it only where we are explicitly discussing the term's provenance. A reader who suspects we have borrowed more authority than we have earned is raising the right objection; this paragraph is our answer, and it concedes most of it.

We are not lawyers. §7 sets out our reading of the applicable provisions. It is a reading, it has not been reviewed by counsel, and it should not be relied on as advice.

The specificity cost is real. Under partial observation the system labels excitement as a possible threat 99.0% of the time. A person who is excited and told to consider whether they are afraid has been mildly misread. We judge this preferable to the converse and do not claim it is costless.

Calibration degrades under severe misspecification. Where the model's assumed structure approximately holds, expected calibration error is 0.044 and the reported uncertainty predicts error with AUC 0.949. Where it does not, ECE rises to 0.221. Reporting a credible set instead of a point label improves the rate at which the truth is contained in the output (78.2% □ 88.0% at $\delta = 0.30$) but does **not** reach nominal coverage: a set claiming 90% mass contains the truth 60.5% [55.5, 65.2] of the time at $\delta = 0.45$. A credible set inherits its posterior's miscalibration; reaching nominal coverage requires conformal calibration against held-out labelled outcomes, which do not exist without a human study.

Correlated evidence. Sequential updating assumes conditional independence, which is frequently false for repeated self-reports. Contributions are keyed by evidence identity so resubmission cannot compound, and temporal decay down-weights clustering, but neither substitutes for a correlated-likelihood model.

Scale. The observation intake recomputes the full belief state per observation and is quadratic (3.8× per doubling; 27 ms at 100 evidence pairs, 1121 ms at 800). This is an engineering limitation, not a conceptual one, and it is the first optimisation target.

9.1 Measured, not asserted: how strong the test suite is

We previously stated that nothing measured the test suite's coverage of the state space. Mutation testing now does. Over 5,114 generated mutants the suite killed 2,319 and 1,549 survived — a **mutation score of 59.9%** — with a further 1,244 mutants in code carrying no covering test, most of it an untested command-line interface that supports no claim here.

The survivors are concentrated in the inference engine, the pattern layer and the invariant checks themselves, and a sampled inspection suggests most are message and label mutations rather than behavioural ones. One was not: mutating an invariant's own identifier survived, meaning nothing asserted that a check reports the invariant it claims to be. That is now tested.

We report the number rather than improving it first, because a paper that argues for checkable claims should publish the check that came back middling.

10. Ethics statement

This research did not involve human or animal subjects. All results derive from simulated signal streams generated from specified processes; no personal data of any kind was collected, processed or analysed.

The system described is a research instrument. It is not a medical device, not a diagnostic instrument, and is not suitable for clinical, educational, employment, credit, insurance, policing or other high-stakes deployment. The prohibited-use list and its enforcement status are given in §8.

The authors note the regulatory position described in §7 and do not assert that the safety exception under Art. 5(1)(f) applies as a settled matter.

Data and code availability. The complete implementation, benchmark suite, experiments and analysis scripts are released under an open licence. The suite is byte-reproducible: two independent runs produce identical output. Every number in this paper is produced by a named script and none was transcribed by hand. *[REPOSITORY URL — TO BE COMPLETED]*

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11. Venue note *(remove before submission)*

Recommended: **AIES** or **FAccT**. The paper's shape — a specific foreseeable harm quantified, a design commitment that removes it, an honest ablation showing the conditions under which the protection fails, and an unresolved regulatory tension — matches what those venues reward, and requires no human subjects. Check current calls for papers before committing; dates move annually.

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Appendix A — Number provenance

Every figure in this paper, with the script that produces it. No number was transcribed by hand.

Figure	Source
All §6.1 rates and intervals	`results.json → b2.conditions[1]` (δ=0.10, avoidance observed)
§6.1 δ=0.00 figures	`results.json → b2.conditions[0]`
§6.1 per-process table	`results.json → b2.conditions[1].per_process`
All §6.2 ablation figures	`results.json → b2.conditions[2]` (avoidance unobserved) vs. Session 2 baseline run
All §6.3 refusal figures	`results.json → b3.results`
§6.4 ambivalence, post-fix (2.5%, 0.0%, 100.0%, 97.6%)	`experiment_ambivalence.json`
§6.4 ambivalence, **pre-fix** (95.5%, 98.8%, 34.0%)	same script run against the pre-fix engine; recorded in `SESSION_08_CLAIM_MATRIX.md` §1. Reproduce by reverting the `_oscillation_gate` factor in `engine._infer_desire`.
§6.4 inquiry regret	`results.json → b4.conditions`
§9 calibration and coverage	`v030_vs_v040.json`, `experiment_d23.json`
§9 complexity	`test_operator_correspondence.py` timing harness

Reproduce with:

```
pytest -q                       # 671 passed, 2 xfailed
python run_benchmarks.py        # writes results.json
python experiment_ambivalence.py
```

```
python experiment_d23.py
python analyze_v030_vs_v040.py
```

Appendix B — Outstanding work with no available substitute

Everything in this appendix was searched for before it was listed. The rule applied was: an item is only recorded here if an active hunt for a public dataset, a free instrument, a third-party service or a methodological substitute came back empty. Items that a search *did* resolve are recorded in the second table, so that a reader can see the difference between what is genuinely blocked and what was merely inconvenient.

B.1 What has no substitute

1. The threat-confound claim requires primary data collection. Approximately forty public datasets were assessed across affective computing, approach-avoidance paradigms, ecological momentary assessment, experimental economics and social conformity. The claim needs four things measured on the same person and the same item: a stated stance, a behavioural choice that can contradict it, an arousal channel time-locked to that choice, and a coercion manipulation. That intersection does not exist publicly. The failure is structural rather than incidental — corpora with good physiology are passive-viewing paradigms in which nothing can contradict a rating, and corpora with real choice lack autonomic recording, self-report, or both. Two published studies used almost exactly the right design and deposited no data. **No workaround exists. Only collection.**

2. No instrument in the battery has been validated in an Egyptian sample. Arabic validations were confirmed for Lebanese, Saudi, Omani, Jordanian, Palestinian, Qatari and Moroccan samples. None is Egyptian. Four candidate Egyptian-journal papers were identified by DOI, but every `*.ekb.eg` host was unreachable from our environment, so their contents are unknown — and Egyptian psychometric papers frequently construct local instruments rather than translating Western ones, so their relevance cannot be assumed. Any use of this battery in Egypt is first-in-population work and is described as such. **Partially resolvable by a colleague with Egyptian institutional access; not resolvable by open search.**

3. Two instrument constructs have no free replacement. A licence hunt found free, published-mapping substitutes for two of the four blocked instruments (see B.2). It did not for the other two. There is no openly licensed eight-theme Eriksonian inventory: the two structurally correct candidates are both permission-gated, and going upstream from one to the other does not escape the gate. The best free material covers two of eight themes and yields single scores rather than subscales. For purpose beyond the Meaning in Life Questionnaire, no free substitute was identified. **Resolvable only by author permission, which is a letter, not a search.**

4. No party outside this project has executed the software. This was the largest risk identified by our own adversarial review, and we searched specifically for mechanisms that would discharge it. The venue route is closed: neither ACM TiiS, nor ACM FAccT, nor AAAI/ACM AIES operates an artifact-evaluation or badging track that an author can enter at submission time. Archival deposit satisfies an availability criterion but verifies no execution. Two replication communities publish work written *by* the replicator and therefore cannot be pointed at an artifact. One community service does exist that matches the need exactly, and submitting to it is recorded in B.2 as an action rather than a limitation — but as of writing it has not been done, and the claim in this paper is that the code is inspectable, not that it has been independently run.

5. Two invariants are syntactic proxies for semantic properties. Invariant X detects verdict phrasing by pattern, and the accompanying question generator is guarded against prescriptive drift by a keyword test. Both raise a floor and neither measures the property. We know of no decision procedure that settles whether a sentence claims authority over a person, and we do not think one follows from more patterns. This is a research problem rather than an unfinished task, and we would rather state it than let a passing check imply otherwise.

B.2 What a search resolved, and how

Recorded so that the list above is read as the residue of a search rather than as its whole.

Was blocked	Resolved by	Cost of the substitution
Values instrument permission-gated, item mapping unreachable	The ESS Human Values Scale, whose items and item-to-value mapping are both published, under CC BY-SA documentation terms plus the instrument author's own non-commercial grant — and which has an already-fielded Arabic version	10 basic values instead of 19 refined, from 2–3 items each
Wisdom instrument permission-gated	The revised Adult Self-Transcendence Inventory, CC BY 4.0, whose original author is a named co-licensor, dimensional, and whose authors argue explicitly against a total score	Construct drift: it measures self-transcendence, and has no analogue of the cognitive dimension
Harness coverage unmeasured	Mutation testing. Reported in §9 as a measured score with its weakest modules named	None; it replaced an admission with a number and a work list
Invariant identity untested	Found *by* the mutation run — mutating an invariant's identifier and name produced surviving mutants — and closed with two tests	None
Distribution-free coverage guarantees blocked on labelled outcomes	Available under the secondary-analysis route, where two of the three corpora supply an item-matched choice that can contradict a stated preference	Applies to the preference model only; the safety claim is untouched
Independent execution	A community code-check service that accepts submissions on a preprint alone, is free, and issues a citable certificate naming the person who ran the code	Certifies execution, not scientific correctness

B.3 What is not in this appendix

Ordinary unfinished work is excluded on purpose: two unimplemented operators recorded as strict `xfail`s, a read path that remains linear per consultation, a candidate belief state for monotonic drift, and the surviving mutants named in §9. None of these is blocked on anything external. They are simply not done, and listing them here would dilute a list whose value is that every entry on it has been searched for and not found.